INCH-POUND

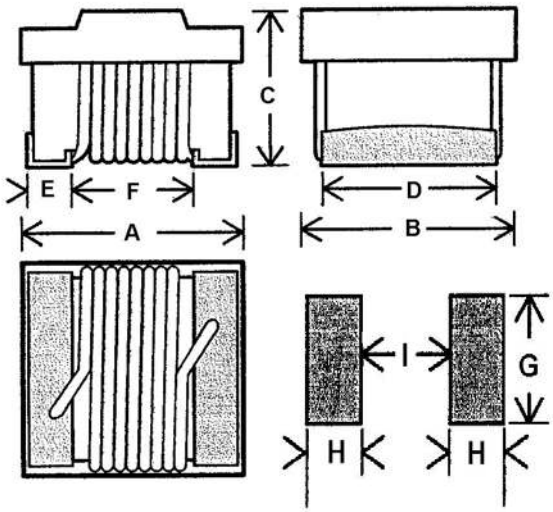
MIL-PRF-83446/36
6 June 2003

PERFORMANCE SPECIFICATION SHEET

COILS, RADIO FREQUENCY, CHIP, OPEN CONSTRUCTION, SURFACE MOUNT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-83446.



LAND PATTERN

	INCHES	MILLIMETERS
A	0.071 Max.	1.80 Max.
B	0.045 Max.	1.14 Max.
C	0.040 Max.	1.02 Max.
D	0.030 (Ref. Only)	0.76 (Ref. Only)
E	0.018 (Ref. Only)	0.44 (Ref. Only)
F	0.040 (Ref. Only)	1.03(Ref. Only)
G	0.040	1.03
H	0.025	0.64
I	0.025	0.64

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration.

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TABLE I. Electrical characteristics (initial).

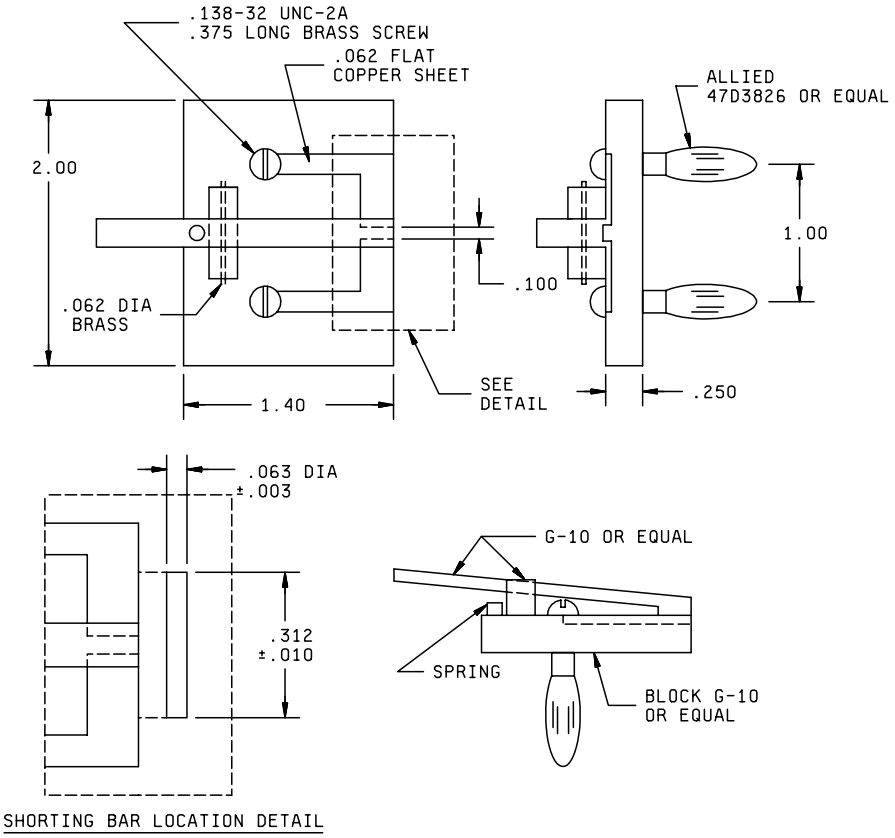
Dash Number	Inductance nH ±10%	Q (min)	Test Frequency (L and Q) MHz	Self Resonant Frequency MHz (min)	DC Resistance Ohms (max)	Current Rating mA (max)
-01	1.6	16	250	6000	0.040	700
-02	1.8	16	250	6000	0.045	700
-03	3.3	22	250	6000	0.080	700
-04	3.6	22	250	5800	0.063	700
-05	3.9	22	250	6000	0.080	700
-06	4.3	22	250	5800	0.063	700
-07	4.7	25	250	5800	0.120	700
-08	5.1	20	250	5800	0.160	700
-09	5.6	20	250	5800	0.170	700
-10	6.8	27	250	5800	0.110	700
-11	7.5	27	250	4800	0.110	700
-12	8.2	27	250	4800	0.110	700
-13	8.7	27	250	4800	0.110	700
-14	9.5	27	250	4800	0.130	700
-15	10	31	250	4800	0.130	700
-16	11	31	250	4000	0.086	700
-17	12	35	250	4000	0.130	700
-18	15	35	250	4000	0.170	700
-19	16	35	250	3300	0.110	700
-20	18	35	250	3100	0.170	700
-21	22	38	250	3000	0.190	700
-22	24	38	250	2800	0.130	700
-23	27	40	250	2800	0.220	600
-24	30	40	250	2800	0.150	600
-25	33	40	250	2300	0.220	600
-26	36	40	250	2300	0.250	600
-27	39	40	250	2200	0.250	600
-28	43	38	200	2000	0.280	600
-29	47	38	200	2000	0.280	600
-30	56	38	200	1900	0.310	600
-31	68	37	200	1700	0.340	600
-32	72	34	150	1700	0.490	400
-33	82	34	150	1700	0.540	400
-34	100	34	150	1400	0.580	400
-35	110	32	150	1350	0.610	300
-36	120	32	150	1300	0.650	300
-37	150	32	100	1300	0.950	280
-38	180	25	100	1200	1.400	250
-39	220	25	100	1200	1.600	250
-40	270	25	100	900	2.100	200

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TABLE II. Electrical characteristics (final).

Inspection group	Allowable percent variation from the initial measurement			
Qualification Inspection:	Inductance	Q	Self-resonant Frequency	DC resistance
Group II Group IV Group V	±5 ±5 ±5	-10 -10 -10	-8 -10 -----	±3 (+.001 ohm) ±3 (+.001 ohm) -----
Quality Conformance Inspection: Group C Subgroup II Subgroup IV	 ±5 ±5	 -10 -10	 -10 -8	 ±2 (+.001 ohm) ±3 (+.001 ohm)

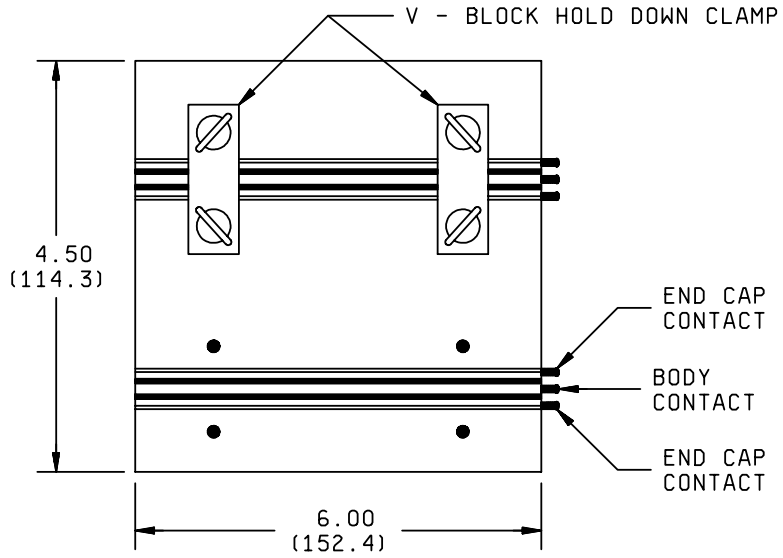
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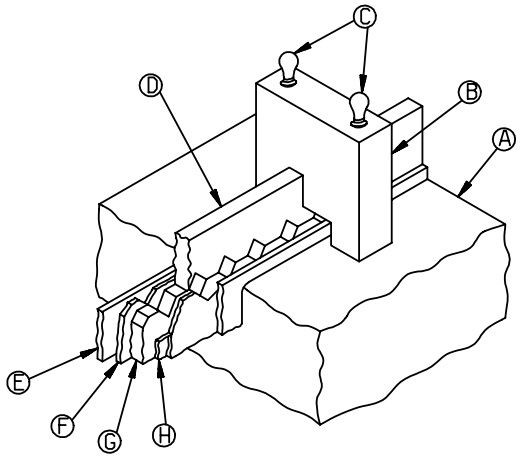


NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Shorting bar material is brass.
4. The total fixture and residual inductance is approximately .049μH (for references only).

FIGURE 2. General test fixture, (TF-397).

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ENLARGED CUT-AWAY OF CLAMP

Legend	
LTR	Description
A	G-10 base
B	Upper V-block hold-down clamp
C	Clamp screws
D	Upper V-block (body contact)
E	End cap contact retaining wall
F	End cap contact V-block (in contact with E)
G	Lower V-block (body contact)
H	G-10 insulators (separating G from F/E)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 3. Test fixture for insulation resistance, dielectric withstanding voltage and barometric pressure.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Bond strength: Force 2 pounds.

Weight: 0.19 gram maximum.

Operating temperature range: -40°C to +125°C.

Temperature rise (at +90°C): +15°C.

Maximum operating temperature: +125°C.

Altitude: 70,000 feet.

Dielectric withstanding voltage: Method 301 of MIL-STD-202, test voltage 500 Vrms for 40 seconds.

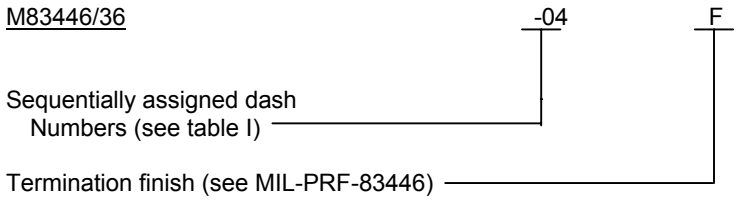
Barometric pressure: Method 105, test condition C, MIL-STD-202, (70,000 feet), test voltage 200 Vrms.

Electrical characteristics (initial): See table I.

Electrical characteristics (final): See table II.

Test fixture: Tests shall be performed using test fixtures shown on figure 2 and 3, or equivalent.

Part or Identifying Number (PIN): The PIN shall be in the following form:



Custodians:
Army - CR
Navy - EC
Air Force - 11

Review activities:
Army - MI
Navy - AS, OS
Air Force - 19

Preparing activity:
DLA - CC

(Project 5950-1148)

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